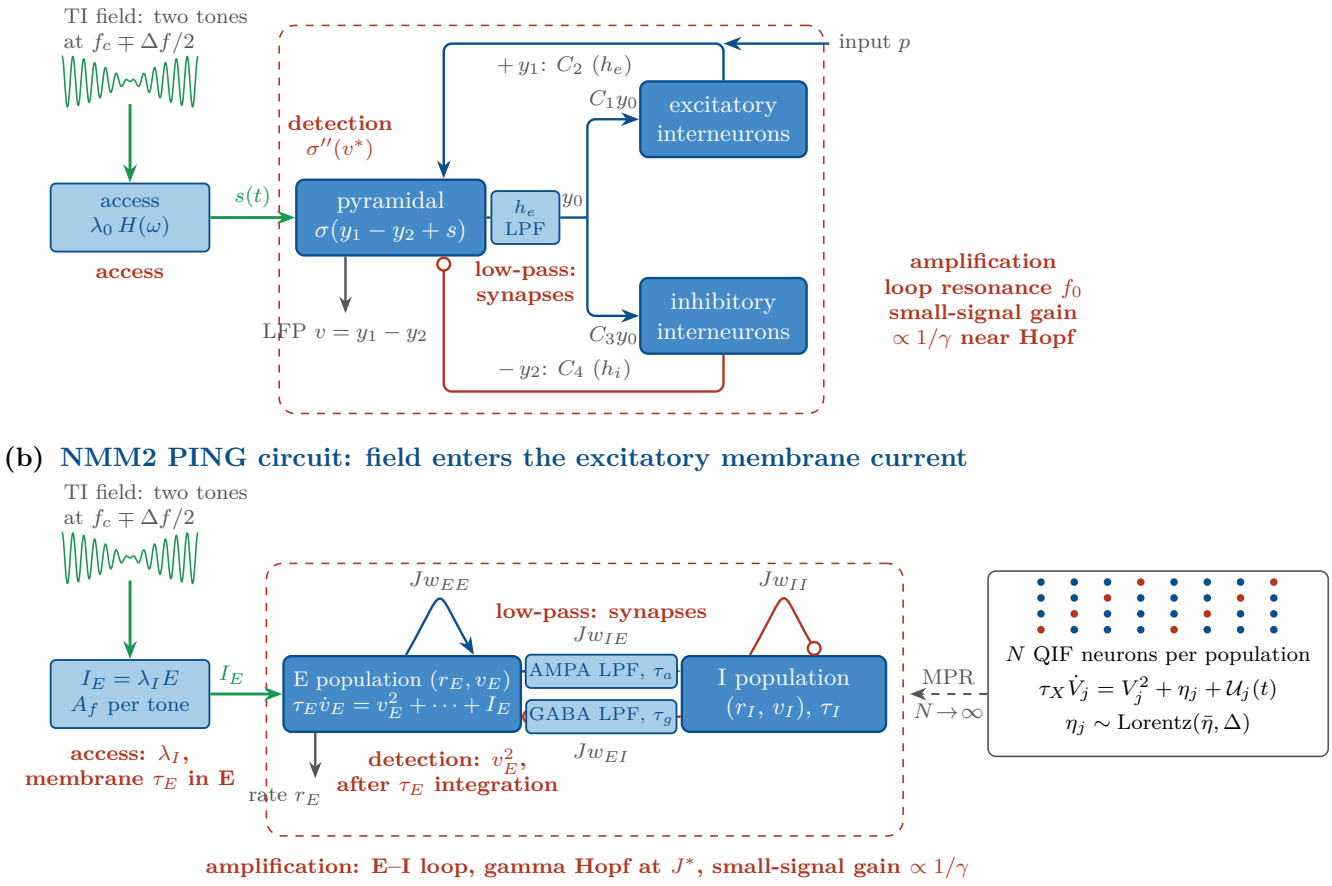

(a) Jansen–Rit column: field enters the pyramidal sigmoid
TI field: two tones at $f_c \mp \Delta f/2$
access $\lambda_0 H(\omega)$
access
$s(t)$
pyramidal $\sigma(y_1 - y_2 + s)$
h_e LPF
y_0
detection $\sigma''(v^*)$
LFP $v = y_1 - y_2$
low-pass: synapses
input p
$+ y_1$: C_2 (h_e)
$C_1 y_0$
excitatory interneurons
inhibitory interneurons
$C_3 y_0$
$- y_2$: C_4 (h_i)
amplification
loop resonance f_0
small-signal gain
$\propto 1/\gamma$ near Hopf
(b) NMM2 PING circuit: field enters the excitatory membrane current
TI field: two tones at $f_c \mp \Delta f/2$
$I_E = \lambda_I E$
A_f per tone
access: λ_I,
membrane τ_E in E
I_E
Jw_{EE}
low-pass: synapses
Jw_{IE}
Jw_{II}
E population (r_E, v_E)
$\tau_E \dot{v}_E = v_E^2 + \cdots + I_E$
AMPA LPF, τ_a
GABA LPF, τ_g
I population (r_I, v_I), τ_I
Jw_{EI}
rate r_E
detection: v_E^2,
after τ_E integration
MPR
$N \to \infty$
N QIF neurons per population
$\tau_X \dot{V}_j = V_j^2 + \eta_j + \mathcal{U}_j(t)$
$\eta_j \sim$ Lorentz$(\bar{\eta}, \Delta)$
amplification: E–I loop, gamma Hopf at J^*, small-signal gain $\propto 1/\gamma$